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Capability Surprise and the Pricing of an Early-Commercial-Stage Foundation-Model Lab: Evidence from Zhipu AI (2513.HK)

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Abstract

In January 2026 Zhipu AI (Knowledge Atlas Technology, 2513.HK) became the first foundation-model laboratory to list publicly anywhere, under Hong Kong’s Chapter 18C listing regime. By 26 June 2026 its shares had risen about seventeen-fold to a market capitalisation near US\$114bn—roughly 570 times its ~US\$200m of expected 2026 revenue—despite a gross margin that has fallen to near zero and negative book equity. We ask what such a firm is worth and how its price forms. A three-scenario DCF, a reverse DCF, a comparables check and a real-options reading place fundamental value at only HK\$11–67 per share (about 1.5% of the market price); the price instead embeds a winner-take-all outcome of more than US\$100bn revenue by 2035. Because conventional earnings-surprise measures are economically uninformative for a deeply loss-making early-stage lab, we replace them with *capability-surprise* events (model releases and benchmark jumps). The market prices capability immediately and discriminately—separating Zhipu from peers on model quality, not sector—with preliminary evidence of post-capability-announcement drift. The valuation gap is best read as capability momentum priced as an option, not a simple bubble.

Keywords: foundation models; Chapter 18C; pre-profit valuation; real options; market efficiency; capability surprise; PEAD/PCAD.

1 Introduction

The first half of 2026 produced an event with no precedent in capital-market history: within forty-eight hours, two Chinese “AI tiger” laboratories—Zhipu AI (2513.HK, 8 January) and MiniMax (00100.HK, 9 January)—became the first foundation-model developers to go public anywhere, listing on the Hong Kong Stock Exchange. Zhipu listed under Chapter 18C, the regime for specialist-technology firms that admits pre-profit issuers. By 26 June 2026 its shares had compounded roughly seventeen-fold to a market capitalisation near US\$114bn on FY2026 expected revenue of about US\$200m—an implied price-to-sales multiple near $570\times$ [Tushare, 2026, Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI), 2025].

This paper poses three questions: **(i)** what is Zhipu worth on fundamentals; **(ii)** what does the re-rating imply—efficient pricing of capability, exuberance, or option value; and **(iii)** is the open-weight, cost-disruption strategy sustainable? Our contribution is methodological: because an early-stage lab has no earnings, the standard earnings-surprise event study [Ball and Brown, 1968, Bernard and Thomas, 1989] has nothing to bite on, so we reframe the information event as a *capability surprise*—a model release or benchmark jump—adapting the surprise-event (*chao yuqi*) design of Guosen Securities Economic Research Institute [2020] to a new asset class. The weight of the paper falls on the valuation (Section 4) and the event study (Section 5); Sections 2–3 give only the context they require.

2 Background: Company, Financials, and Listing

Business model. Founded in 2019 out of Tsinghua University’s Knowledge Engineering Group, Zhipu develops the GLM (“General Language Model”) family and monetises through usage-based API access, enterprise subscriptions, and private/on-premise deployment. Its strategic

choice is an *open-weight funnel*: flagship models are released with open weights and aggressive token pricing, converting developer mindshare into a usage flywheel via the Z.ai and Z.code platforms.

Financials (full statements in Appendix A). Three facts dominate. (1) Revenue is explosive but tiny: RMB 312.4m (2024) $\rightarrow$ RMB 724.3m (2025, +132%), i.e. roughly US\$44m then US\$102m. (2) The gross margin has *collapsed* from 76.1% (2022, a normal software profile) to 3.4% (2024) and to -0.4% in H1 2025, as the firm bought share through compute-heavy, under-priced workloads. (3) Losses and leverage are deepening: the 2025 net loss was about RMB 4.70bn (+237%), and at 30 June 2025 the group reported net liabilities of RMB 6.15bn against cash of only RMB 190.9m [Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI), 2025]. Zhipu is therefore *early-commercial-stage and pre-profit*, with negative book equity—so ratio analysis is of limited use and the informative metrics are growth, the gross-margin path, and cash runway.

Listing and market reaction. Zhipu priced at HK\$116.20, raising about HK\$4.3bn with HK\$3bn of cornerstone commitments, at an implied valuation near US\$6.7–7.4bn [Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI), 2025]. The listing is strategically remarkable on three counts: Chapter 18C gave a loss-making lab permanent public capital years before profitability; being the world’s first listed foundation-model company conferred a certification premium; and the stock then re-rated to about HK\$2,046 by 26 June 2026 ($\approx +1,660\%$ versus IPO; realised annualised volatility near 193%; Figure 4). MiniMax, listing one day later at HK\$165, doubled on debut and then *de-rated* by roughly 60% from its peak—a divergence we exploit in Section 5.

3 Industry, Macro, and Strategy

Structure and strategy. In Porter terms the foundation-model layer is harsh: supplier power (advanced compute) is high and politically rationed, buyer switching costs are low because open weights commoditise the layer, and the substitutes are rival open-weight models. Zhipu’s answer is leaderboard dominance through *cost disruption*: a mixture-of-experts GLM-5 line (GLM-4.6: $\approx 355\text{B}$ parameters, 32B active, 200K context) that topped coding benchmarks—GLM-5.1 ranked #1 on SWE-Bench Pro at 58.4%, and GLM-5.2 (June 2026) shipped with MIT open weights and a one-million-token context—priced below closed competitors, now paired with nascent pricing power (8–17% API price rises with each GLM-5.x release) [Zhipu AI / Z.ai, 2026]. This cadence defines the event taxonomy of Section 5. The main weakness is self-inflicted: the same cadence commoditises its own product, while the near-zero gross margin and compute dependence cap monetisation.

The Hong Kong AI-IPO landscape. The 2026 listing wave spans two business models. Zhipu and MiniMax are general-purpose foundation-model laboratories; a third entrant, **Wenge AI** (中科闻歌, 01956.HK), listed on 26 June 2026 at HK\$60.70 and closed its first day near HK\$111.7 (+84%), at about HK\$10.5bn ($\sim$ US\$1.3bn) [Wenge Technology Co., Ltd. (中科闻歌), 2026]. Founded in 2017 by a Chinese Academy of Sciences (Institute of Automation) team, Wenge ranked first by revenue in China’s enterprise decision-intelligence large-model market in 2025 ($\sim 10.2\%$ share), selling a DIOS decision-intelligence operating system and the Decitron and Yayi models mainly to government, media, and enterprise clients. *Recent Hong Kong AI listings include foundation-model laboratories such as Zhipu AI and MiniMax, as well as enterprise-focused AI platforms such as Wenge AI; however, Wenge is better classified as a decision-intelligence solution provider than as a direct frontier-model comparable.* It therefore enters this paper as cross-sectional context, not as a valuation comparable (Section 4.4).

Macro and flows. Chinese AI is a policy priority (state/SOE demand); U.S. compute-export

controls force the architectural efficiency that doubles as Zhipu’s cost edge; and Hong Kong’s market structure sets the marginal buyer—Zhipu joined the Hang Seng Tech Index (5 June) and Southbound Stock Connect (8 June 2026), drawing about HK\$920m of net southbound buying on its first eligible day [Hang Seng Indexes Company Limited, 2026]. These flow events matter for Section 5, where we separate them from capability events.

4 Valuation

4.1 Cost of capital

We estimate the cost of equity with the CAPM. Given Zhipu’s extremely short post-IPO trading history, an estimated regression beta would be unstable and dominated by IPO- and event-driven shocks; we therefore construct a bottom-up beta from global AI-software comparables, giving $\beta \approx 1.6$. With a risk-free rate of 4.0% and an equity risk premium of 6.0%, $K_e = 4.0\% + 1.6 \times 6.0\% = 13.6\%$; as the firm is effectively all-equity, $WACC \approx 13.5\%$. Throughout we use HK\$7.8/US\$, RMB 7.1/US\$, ≈ 435 m shares, and net cash of about **US\$0.55bn**—net IPO proceeds of $\sim$ HK\$4.3bn ($\sim$ US\$0.55bn) plus pre-IPO cash of $\sim$ US\$0.03bn. The pre-IPO net-liability position is dominated by redeemable preferred shares that convert to ordinary equity on listing, so it is not post-IPO debt; treating any residual as debt would only lower equity value and reinforce our conclusion.

4.2 Scenario DCF

We project FCFF over 2026–2035 from a US\$200m base, with revenue growth fading from a scenario-specific start to a mature rate, operating margin ramping from -30% to a terminal level, a 15% tax rate, and reinvestment via a sales-to-capital ratio of 1.6–2.0; terminal value uses $g = 4\%$. All cells are live in the companion workbook (`model/valuation_model.xlsx`); the full Base-case chain is in Appendix B, and headline results in Table 1.

Table 1: Three-scenario DCF (USD unless noted; per-share in HKD). Market row at 2026-06-26.

Scenario	Rev. CAGR '26–35	Term. margin	Equity value	Per share (HK\$)
Bear ($p = 0.35$)	21%	18%	US\$0.6B	11
Base ($p = 0.45$)	31%	28%	US\$1.6B	29
Bull ($p = 0.20$)	43%	35%	US\$3.7B	67
Probability-weighted	—	—	US\$1.7B	30
<i>Market (2026-06-26)</i>	—	—	<i>US\$114B</i>	<i>2,046</i>

The probability-weighted intrinsic value is about HK\$30 per share—roughly **1.5% of the market price**. No reasonable point estimate of fundamentals approaches the quote.

4.3 Reverse DCF: what the price requires

We invert the model to ask what would justify the US\$114bn equity value. Holding WACC at 13.5% and a 40% terminal margin, the price requires revenue of **more than US\$100bn by 2035**—an annual growth rate near 105% over 2026–2035 (nine growth intervals), several hundred times FY2026E. The exact figure is sensitive to the reinvestment assumption, but the qualitative message is robust: the price embeds a hyperscaler-tier outcome reserved for a handful of global winners.

4.4 Relative valuation

On FY2026E revenue Zhipu trades near **570 $\times$ sales**. For context, leading private frontier labs (OpenAI, Anthropic) have transacted around 30–40 $\times$ trailing revenue, and MiniMax—the most appropriate direct comparable—trades far lower than Zhipu on the same basis. We deliberately *exclude* Wenge AI from the multiple comparison: its government/enterprise project-delivery model is not comparable to a frontier model-API and developer-ecosystem business. Zhipu is thus priced roughly an order of magnitude above even the richest private peers—consistent with the reverse-DCF reading rather than with comparable multiples (Figure 1).

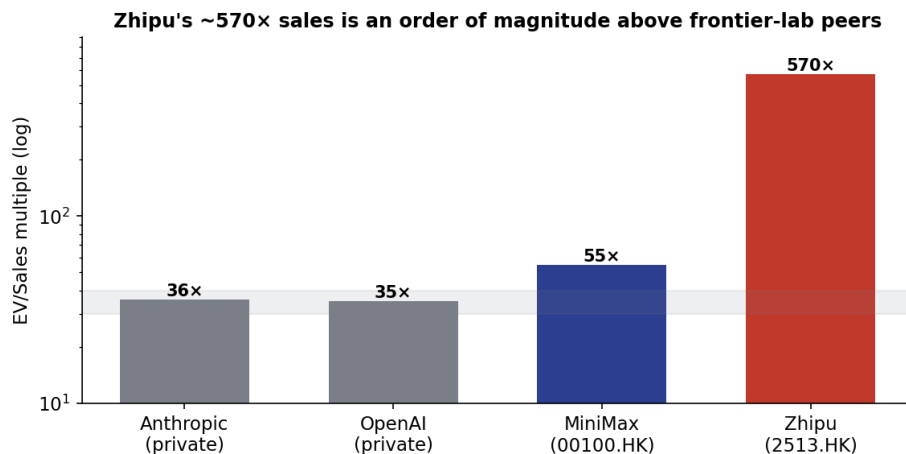


Figure 1: EV/Sales multiples (log scale). Zhipu's $\approx 570\times$ is roughly an order of magnitude above the 30–40 $\times$ at which leading private frontier labs have transacted; even MiniMax sits far below. Private-lab multiples are illustrative.

4.5 A real-options reading

The reconciliation is that Zhipu is best valued not as a discounted cash-flow stream but as a *call option on a winner-take-all AGI outcome* [Black and Scholes, 1973, Damodaran, 2018]. The DCF base case is the option's modest intrinsic value; the bulk of the market price is time value on a fat right tail—high volatility ($\approx 193\%$ annualised) and a long horizon both inflate that value. This separates *fundamental value* from *option value* and explains why the stock can rationally trade far above any DCF point estimate while remaining acutely sensitive to capability news—the subject of Section 5.

A practitioner lens: reflexivity. Reading the expectations embedded in the price—the reverse-DCF move—echoes the reverse-investing framework of the fund manager Feng Liu, who applies Soros's *reflexivity* to equity pricing [Feng, 2017]: a high valuation is not only an output but an input, as cheap equity capital funds the compute and talent that improve the models, which in turn lift the valuation. For Zhipu this loop is unusually direct—and unusually fragile, since Feng Liu also warns that firms with limitless financing needs and no self-generated cash flow (unlike a Moutai-type internal compounder) depend on the loop staying intact. If capability surprises stop arriving, the reflexive support for the option value can unwind as fast as it built.

4.6 Sensitivity and synthesis

The conclusion is not an artefact of any single assumption. Figure 2 tornadoes the Base case against its main drivers: even jointly bullish settings (WACC 12%, terminal margin 33%, 72% start growth) lift value only to the mid-HK\$30s—still about two orders of magnitude below the

market. Figure 3 draws the whole picture together as a valuation football field: scenario DCF (HK\$11–67) and a comparables cross-check (HK\$108–143 at 30–40 $\times$ sales) both sit far to the left of the HK\$2,046 quote, a gap of roughly 60–70 $\times$. The market is not pricing a different point on the same distribution; it is pricing a different distribution—the right-tail option of Section 4.

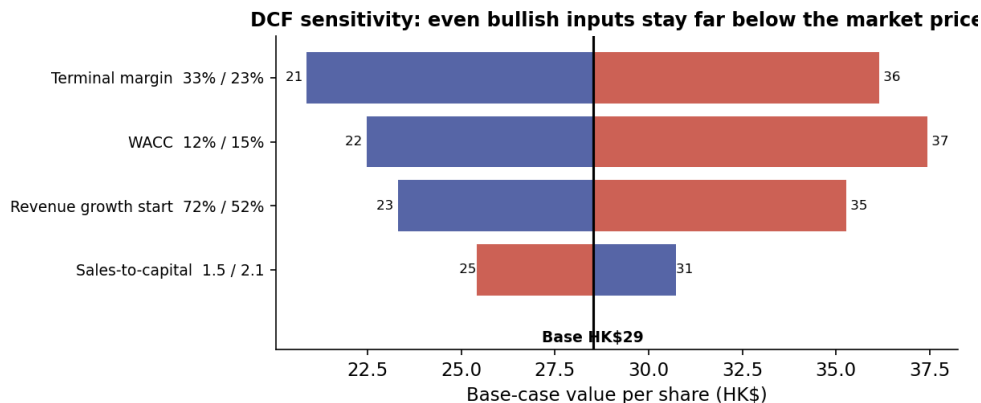


Figure 2: DCF sensitivity (Base case, per share). Bars span each driver’s low/high setting around the HK\$29 base; no single input moves value near the market price.

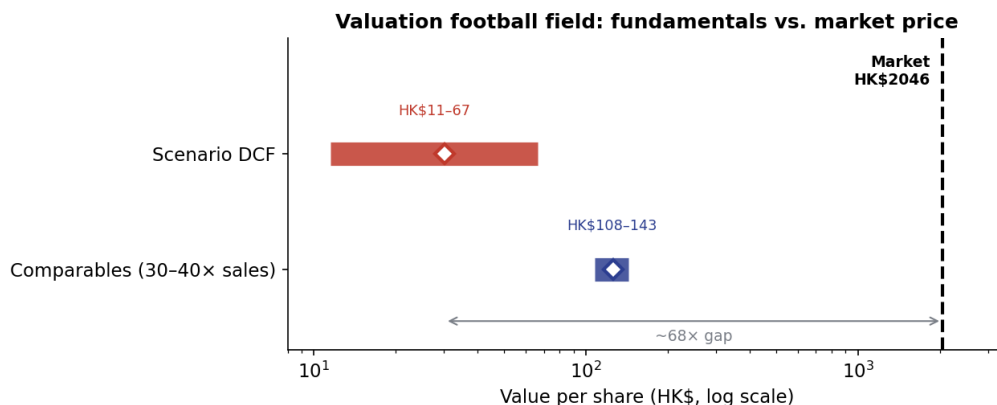


Figure 3: Valuation football field (log scale). Fundamental approaches cluster two orders of magnitude below the market price; the diamonds mark the probability-weighted DCF and the comparables midpoint.

5 Capability Surprise: An Event Study

Hypothesis. If the firm is priced on its capability trajectory, then capability information should move the price. Under semi-strong efficiency [Fama, 1970] the reaction should be immediate and complete; any systematic *post-capability-announcement drift* (PCAD)—the analogue of the post-earnings drift of Ball and Brown [1968], Bernard and Thomas [1989]—would indicate inefficiency.

Method. Event dates are identified *independently* from official model-release announcements and technical documentation [Zhipu AI / Z.ai, 2026], not from the price series. On daily prices [Tushare, 2026] we compute mean-adjusted abnormal returns and CARs over three windows—leakage $[-5, -1]$, reaction $[0, +1]$, and drift $[+2, +10]$ —following standard event-study practice [MacKinlay, 1997] and the surprise-event design of Guosen Securities Economic Research Institute [2020]. For each event the expected daily return is Zhipu’s mean return over the pre-event

estimation window $[-20, -6]$, and the abnormal return is $AR_t = R_t - \bar{R}_{[-20, -6]}$. Figure 5 then serves as a *descriptive cross-check*: several of the largest positive-return days coincide with these independently dated capability events.

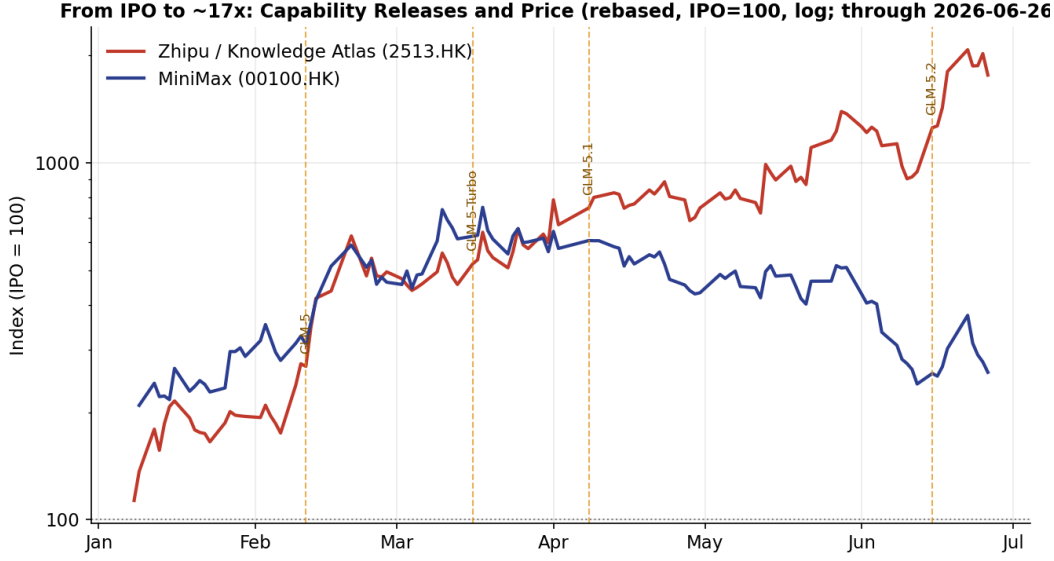


Figure 4: Rebased prices (IPO = 100, log scale, through 2026-06-26) with GLM release markers. Same sector, divergent paths: Zhipu re-rates on successive SOTA releases while MiniMax de-rates from its peak.

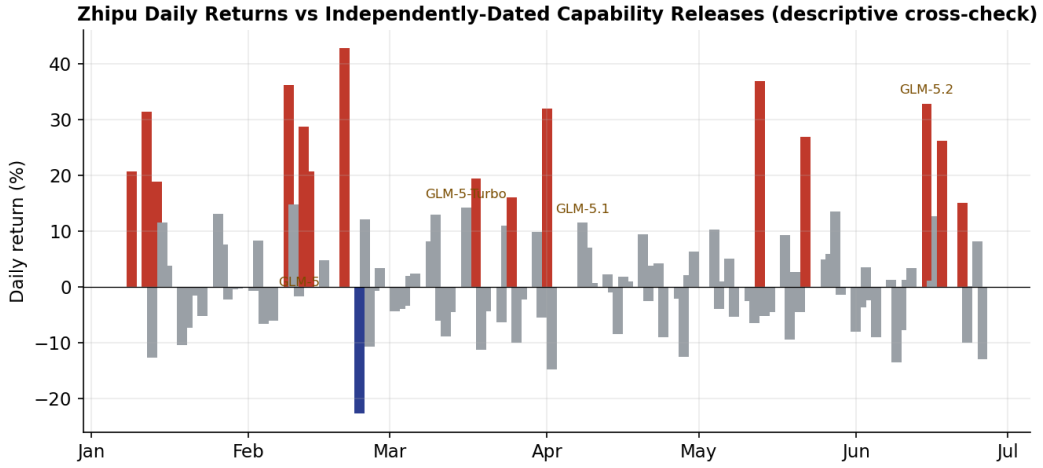


Figure 5: Descriptive cross-check: several of Zhipu’s largest up-days coincide with independently dated GLM releases. (Event dates are taken from official announcements, not from this chart.)

Results. Averaged across the four GLM events, the reaction window is +18.7% (the market prices capability, fast) and the drift averages only +4.8% but is strongly *bimodal* (Table 2). The drift sign depends on the *type* of release: a genuine SOTA leap (GLM-5.2) continues upward, whereas incremental upgrades (GLM-5-Turbo, GLM-5.1) reverse. MiniMax’s M2.7 drew a negligible reaction and a sharp subsequent de-rating (−49%, partly the unwinding of its own debut bubble), consistent with *capability surprise, not sector membership, driving the cross-section* (Figures 6–7).

Table 2: Cumulative abnormal returns around capability events (mean-adjusted, %).

Event	Day 0	Reaction [0, +1]	Drift [+2, +10]	Reading
GLM-5	2026-02-11	+22.5	+24.7	under-reaction
GLM-5-Turbo	2026-03-16	+7.7	−19.3	reversal
GLM-5.1	2026-04-08	+13.8	−14.2	over-reaction
GLM-5.2	2026-06-15	+30.8	+28.2 [†]	strong under-reaction
<i>MiniMax M2.7</i>	2026-03-18	−5.5	−49.1	de-rate
Average (4 GLM)		+18.7	+4.8[†]	

Mean-adjusted: $AR_t = R_t - \bar{R}_{[-20, -6]}$. [†] GLM-5.2's [+2, +10] window is *truncated* (only 7 of 9 trading days fall on/before the 2026-06-26 cut-off), so its drift—and the 4-event average drift—are provisional.

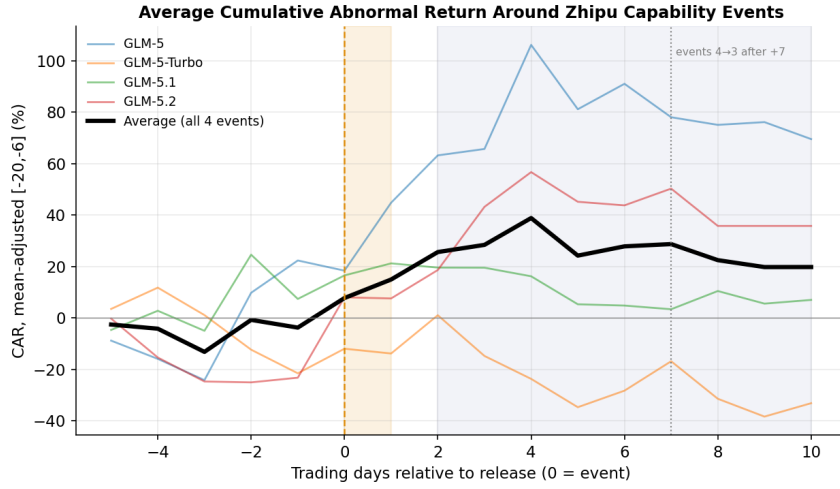


Figure 6: Average CAR in event time across the four GLM events (mean-adjusted, $[-20, -6]$ estimation window). The average is plotted only where all four events have data; the available count falls from four to three after day +7 (GLM-5.2 window truncated at the 2026-06-26 cut-off).

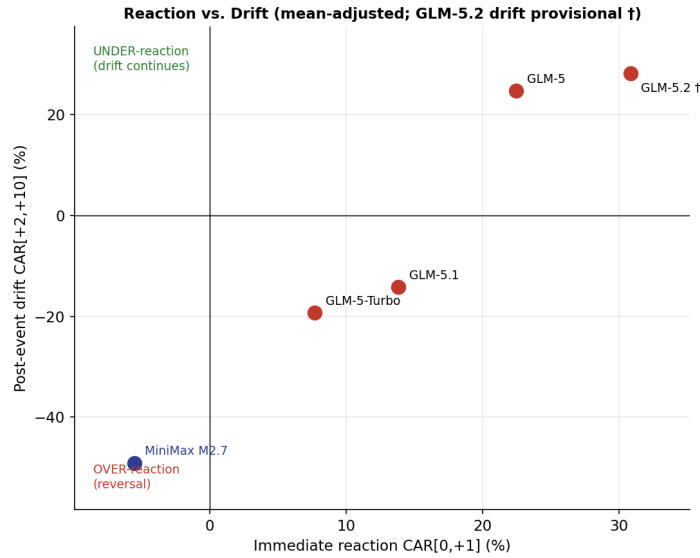


Figure 7: Reaction versus drift. SOTA leaps sit in the under-reaction quadrant; incremental upgrades in the over-reaction quadrant.

5.1 Not every spike is capability: flow events

Two of the largest blind-detected up-days do *not* map to a model release: +43% on 2026-02-20 and +37% on 2026-05-13. Both coincide with *index/flow* catalysts—rising expectations of Hang Seng Tech inclusion and Stock Connect access (realised 5 and 8 June). Separating capability events from flow events matters: the CARs above are driven by the former, while the latter explain part of the raw volatility and warn against attributing every move to model quality.

5.2 Robustness: peer-adjusted abnormal returns

Because Zhipu and MiniMax share sector and macro shocks, we re-run the study with *peer-adjusted* returns (benchmark = MiniMax’s same-day return). Table 3 shows the reaction result is highly robust (+18.7% vs +18.5%), while the average drift *strengthens* (+4.8% $\rightarrow$ +19.2%): once the falling sector is removed, all four events drift positively, and GLM-5.1’s apparent reversal flips to continuation (−14.2% $\rightarrow$ +12.9%). The under-reaction pattern is reinforced, not weakened, by peer adjustment.

Table 3: Robustness: mean-adjusted vs peer-adjusted CAR (%).

Event	Mean-adjusted		Peer-adjusted	
	React [0, 1]	Drift [2, 10]	React [0, 1]	Drift [2, 10]
GLM-5	+22.5	+24.7	+17.2	+13.3
GLM-5-Turbo	+7.7	−19.3	+14.6	+19.2
GLM-5.1	+13.8	−14.2	+13.4	+12.9
GLM-5.2	+30.8	+28.2 [†]	+28.7	+31.3 [†]
Average	+18.7	+4.8 [†]	+18.5	+19.2 [†]

[†] GLM-5.2 drift window truncated (7/9 days through 2026-06-26); the GLM-5.2 rows and the average drifts are provisional.

Honesty box. This is *preliminary, diagnostic* evidence consistent with PCAD, not a proven anomaly: $n = 4$ single-firm events (plus one peer) over five months; GLM-5.2’s drift window is truncated; and drift windows can overlap a fast release cadence. No statistical significance is claimed—four events make an interesting undergraduate case, not a general law. The natural extensions are a wider, multi-lab event panel (adding MiniMax and Wenge releases) and an NLP-scored surprise magnitude.

6 Conclusion and Implications

A disciplined valuation cannot reconcile Zhipu’s fundamentals (HK\$11–67 per share) with its market price (HK\$2,046 on 26 June 2026). The reverse DCF shows why: the price embeds a winner-take-all, US\$100bn-revenue outcome, best expressed as a call option on frontier AI rather than a cash-flow stream. The event study shows *how* that price is discovered—through capability surprises priced immediately and discriminately, with preliminary evidence of under-reaction (a PCAD that survives peer adjustment), while separate index/flow events drive part of the rest. The verdict is neither “efficient” nor “bubble” but **capability momentum priced as an option**. For investors, this argues for trading the capability calendar with attention to release *type*; for founders, Chapter 18C has made capability milestones the primary currency of value; for policymakers, compute access is now a direct input to capital formation. The main

limitations are the small event sample and a near-zero gross margin whose future path is highly uncertain.

A Financial Statements Summary

From the Chapter 18C prospectus and the H1 2025 interim disclosure (RMB; USD at ≈ 7.1).

Table 4: Income-statement summary (RMB millions unless noted).

	FY2022	FY2023	FY2024	FY2025
Revenue	—	≈ 120	312.4	724.3
growth	—	—	—	+132%
Gross margin	76.1%	—	3.4%	$\approx 0\%$ (H1: -0.4%)
Net loss	—	—	2,958.0	$\approx 4,700$
loss growth	—	—	—	+237%

Table 5: Balance-sheet highlights (RMB millions, at 30 June 2025).

Item	Amount
Cash and cash equivalents	190.9
Net liabilities (negative equity)	$-6,150.8$
Net current liabilities	$-9,564.8$
Shares outstanding (post-IPO, approx.)	$\approx 435\text{m}$
IPO price / gross proceeds	HK\$116.20 / $\approx$ HK\$4.3bn

The gross-margin collapse from 76% to near zero is the central operating fact: Zhipu is converting a high-margin software model into a compute-subsidised, share-grab model. Combined with negative equity, it is why the valuation in Section 4 must rest on a future-option logic rather than on current fundamentals.

B Base-Case DCF Projection

Base-case chain (US\$m unless noted), reproducing the live workbook: WACC = 13.5%, tax = 15%, terminal $g = 4\%$, sales-to-capital = 1.8, ≈ 435 m shares, HK\$7.8/US\$. Revenue grows from US\$200m fading $62\% \rightarrow 8\%$; operating margin ramps $-30\% \rightarrow 28\%$.

Table 6: Base-case projection (US\$ millions unless noted).

Year	Revenue	Growth	EBIT margin	NOPAT	Reinvest	FCFF
2026	200	62%	-30%	-51	43	-94
2027	312	56%	-24%	-62	62	-125
2028	468	50%	-17%	-68	87	-155
2029	674	44%	-11%	-61	114	-176
2030	930	38%	-4%	-33	142	-176
2031	1,228	32%	2%	23	165	-142
2032	1,547	26%	9%	114	177	-63
2033	1,856	20%	15%	238	172	67
2034	2,116	14%	22%	388	144	243
2035	2,285	8%	28%	544	94	450

Terminal value (2035) = $450 \times 1.04 / (0.135 - 0.04)$ = US\$4,925m; PV of TV = US\$1,388m.
Enterprise value = US\$1,040m; + net cash US\$550m = equity US\$1,590m = **HK\$29/share**.

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